JOELLE KHALIL

Senior Computer Vision & Embedded AI Engineer

+961 76 44 71 36 · +33 6 83 85 42 88 · joellekhalil97@gmail.com · linkedin.com/in/joellekhalil1997 · Lebanon | Seeking relocation: France · Germany · Switzerland · Luxembourg

PROFESSIONAL SUMMARY

Founding-team engineer at an event-based vision startup with 5+ years designing and shipping computer vision, visual odometry, SLAM, and edge AI systems from prototype to production. Combines rare expertise in event-based sensors, FPGA pipelines, and embedded C/C++ with hands-on PyTorch/TensorFlow model training, large-scale dataset curation, and Linux-based edge deployment. Led international R&D collaborations across three universities, mentored 19 engineers and students, and builds end-to-end embedded AI systems as a hands-on personal practice. Trilingual: English C2, French C2, Arabic C2. Eligible for EU Blue Card with employer sponsorship.

TECHNICAL SKILLS

Computer Vision & AI: Computer Vision, Event-Based Vision, Visual SLAM, Visual Odometry, Object Detection, Object Tracking, Pose Estimation, YOLO, MediaPipe, TinyML, Neural Networks, Deep Learning, Machine Learning

ML Frameworks & Tools: PyTorch, TensorFlow, Edge Impulse, X-CUBE-AI, Dataset Training, Model Benchmarking, Model Optimisation

Embedded & Edge AI: Embedded C, Embedded C++, RTOS, ARM Cortex-M, STM32, Microchip PIC, Edge Inference, TinyML Deployment, Embedded Linux

FPGA & Hardware: VHDL, Verilog, Vivado, Quartus, ModelSim, MIPI Camera Interface, FPGA Sensor Pipelines, SFP+, QSFP, ITLA, CMIS

Software & Platforms: Python, C++, Bash, CMake, Qt, Linux, PetaLinux, Git, n8n, Ollama, Streamlit, Hugging Face, Gemini AI, Termux

Security & Networking: MACsec, AES-CMAC, KDF, Cryptographic Key Management, OTN, OIF Standards

PROFESSIONAL EXPERIENCE

Founding Member / Lead Research Engineer · **Oculi** | Lebanon / Rochester, USA Jun 2020 - Oct 2023

Started as Application Engineer; promoted to Lead Research Engineer. Event-based vision and edge AI.

- Founding team member: contributed to core product architecture, sensor evaluation workflows, and AI-driven perception systems from day one
- Led 3 international R&D collaborations (University of Zurich, AUB, LAU) and mentored 19 people: 8 final-year engineering students across 3 projects, 1 master's thesis student, and 10 interns
- Designed and benchmarked real-time computer vision and machine learning pipelines on event-based sensors (Dynamic Vision Sensors, DVS) capable of up to 1,000 fps, enabling visual odometry, stereo SLAM, and high-speed object tracking at latencies not achievable on general-purpose hardware
- Built and curated large-scale training datasets through multi-month proprietary data collection campaigns combined with public benchmarks (KITTI, Berkeley DeepDrive); trained and deployed YOLO and MediaPipe models for object detection, pose estimation, and landmark tracking using PyTorch and TensorFlow
- Designed and implemented FPGA-based sensor data acquisition pipelines (VHDL, MIPI camera interface) for real-time vision processing, achieving edge inference performance beyond the capability of general-purpose CPU or GPU platforms
- Worked with a diverse sensor suite including HD cameras, DVS event cameras, LiDAR, stereo vision cameras, and thermal sensors; collected and calibrated multi-sensor datasets for computer vision and machine learning pipelines
- Cross-compiled and deployed optimised C++ computer vision applications on Linux and PetaLinux using CMake for embedded, resource-constrained targets
- Built VR and AR vision demos showcasing real-time sensor fusion capabilities to partners and investors
- Achieved 4th place out of 29 global teams at TinyML Hackathon 2023, building a pedestrian detection system for Vision Zero, City of San Jose

Senior Embedded Systems Engineer · **Beyond Silicon Solutions** | Beirut, Lebanon Nov 2023 - Present

Deepening embedded systems expertise at the firmware level, complementing a strong computer vision and AI background to cover the full stack from vision algorithms to bare-metal hardware.

- Developed embedded firmware for proprietary high-speed optical transceivers (SFP+ and QSFP) including OTN management, MACsec secure communication, and OIF-compliant management interfaces (CMIS, SFF-8436)
- Implemented cryptographic modules including AES-CMAC, KDF, and RFC 3394, and designed host-module communication architectures across 2 hardware platforms
- Conducted system validation and 10G optical testing; provided on-site technical support at customer facilities in the Netherlands

FPGA Design Engineer · **IPG Photonics** | Beirut, Lebanon April 2019 - May 2020

- Developed embedded software and system-level simulations for 10G optical communication systems on Intel FPGA platforms; transmitted and received 10G OTU2 frames
- Designed and validated I2C communication modules; automated simulation runs and coverage analysis with TCL scripting

TECHNICAL PROJECTS

Embedded AI Currency Classifier - STM32 Neural Network Inference

- Trained a computer vision model on USD bill images (6 denominations: $1 to $100) using Edge Impulse and deployed onto STM32H755 via X-CUBE-AI, with AI inference running entirely on the microcontroller
- Achieved 58ms inference time within 268KB flash and 393KB peak RAM on a $15 production chip; full pipeline: Termux image capture, n8n routing, Python bridge, STM32 inference, Lovable live dashboard

Behavioral State Monitoring System

- Real-time behavioural tracking pipeline using Samsung Watch events and MacroDroid triggers; time-series metrics in a Streamlit dashboard with a locally-hosted large language model (Ollama) for privacy-first adaptive insights

Smart Image Collector and Analyser

- Automated image capture from mobile devices and Meta smart glasses via Termux; n8n pipeline for GPS extraction, object detection, and large language model summaries delivered by email

AI Market Intelligence System

- Real-time tracking of cryptocurrency (BTC, ETH, SOL) and commodities via APIs; momentum-based signals with confidence scoring; Gemini AI market commentary with alerts via email and Telegram

EDUCATION

Bachelor of Engineering - Electrical and Electronics Engineering · **Holy Spirit University of Kaslik (USEK)** | Lebanon Sept 2015 - Jun 2020

- GPA: 91.19 / 100 | Dean's List 2017-2020 (all years) | EQF Level 6
- Final Year Project: OTU2 Tester, developed in collaboration with IPG Photonics

CERTIFICATIONS

- Tiny Machine Learning (TinyML) - HarvardX Series (2021) - Applied directly to edge inference pipelines at Oculi
- Cortex-M: Modular Embedded Systems Design - Udemy (2024)
- Semiconductor Design 101 - Purdue University (2025)
- Exploring Linux Build Systems (Buildroot)
- Claude Code Masterclass: Code 5x Faster with Agentic AI - Udemy

LANGUAGES AND ADDITIONAL INFORMATION

English C2 | French C2 | Arabic C2 (native) | Work authorisation: Eligible for EU Blue Card (employer sponsorship required) | International experience: Europe, USA, Middle East